Digital Data Transmission 2026–27

Universitat Pompeu Fabra Barcelona

Seminar 4. Information through discrete channels

A memoryless channel has input $X \in \{0,1\}$ and output $Y \in \{0,1,\star\}$, where $\star$ denotes an erasure. The channel law is

$W(y\|x)$	$y=0$	$y=\star$	$y=1$
$x=0$	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{4}$
$x=1$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{2}$

.

The input is equiprobable in questions 1 to 5 and in question 7. In question 6, $\Pr(X=1)=q$ may vary over $[0,1]$. All entropies are in bits and mutual information is in bits per channel use. For interpretation, $\log_2 3 \simeq 1.585$. We use $0\log_2 0 = 0$.

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1. What is the joint probability $\Pr(X=1, Y=\star)$?

(a) $\frac{1}{4}$
(b) $\frac{1}{8}$ $\star$
(c) $\frac{1}{2}$

Solution: By the product rule in Section 3.2 of the notes, we obtain

$$\Pr(X=1, Y=\star) = \tfrac{1}{2}\cdot\tfrac{1}{4} = \tfrac{1}{8}.$$

The channel-table entry $\frac{1}{4}$ is conditional on $X=1$. The answer is (b).

2. What is the output distribution, in the order $\{0,\star,1\}$?

(a) $\{\frac{3}{8},\frac{1}{4},\frac{3}{8}\}$ $\star$
(b) $\{\frac{1}{4},\frac{1}{2},\frac{1}{4}\}$
(c) $\{\frac{1}{2},\frac{1}{4},\frac{1}{4}\}$

Solution: By the output-law formula in Section 3.2, we sum over both inputs,

$$\Pr(Y=0) = \tfrac{1}{2}\cdot\tfrac{1}{2} + \tfrac{1}{2}\cdot\tfrac{1}{4} = \tfrac{3}{8}.$$

Symmetry gives $\Pr(Y=1)=\frac{3}{8}$. Both inputs have erasure probability $\frac{1}{4}$, so $\Pr(Y=\star)=\frac{1}{4}$. The answer is (a).

3. What is the output entropy $H(Y)$?

(a) $\frac{3}{2}$
(b) $3-\log_2 3$
(c) $\frac{11}{4}-\frac{3}{4}\log_2 3$ $\star$

Solution: Using the entropy definition in Section 3.1 and question 2, we find

$$\begin{aligned} H(Y) &= -2\cdot\tfrac{3}{8}\log_2\tfrac{3}{8} - \tfrac{1}{4}\log_2\tfrac{1}{4} \\ &= \tfrac{3}{4}(3-\log_2 3) + \tfrac{1}{2} \\ &= \tfrac{11}{4} - \tfrac{3}{4}\log_2 3. \end{aligned}$$

The three outputs are not equiprobable. The answer is (c).

4. What is the conditional entropy $H(Y|X)$?

(a) 1
(b) $\frac{3}{2}$ $\star$
(c) $\frac{11}{4}-\frac{3}{4}\log_2 3$

Solution: Section 3.1 averages the entropies of the channel rows. Each row contains the probabilities $\frac{1}{2},\frac{1}{4},\frac{1}{4}$, whose entropy is

$$-\tfrac{1}{2}\log_2\tfrac{1}{2} - 2\cdot\tfrac{1}{4}\log_2\tfrac{1}{4} = \tfrac{3}{2}.$$

Since both row entropies equal $\frac{3}{2}$, their average is $H(Y|X)=\frac{3}{2}$, whatever the input distribution. The answer is (b).

5. What is the mutual information $I(X;Y)$?

(a) $\frac{5}{4}-\frac{3}{4}\log_2 3$ $\star$
(b) $\frac{3}{4}$
(c) $\frac{11}{4}-\frac{3}{4}\log_2 3$

Solution: By the mutual-information identity in Section 3.2, we subtract

$$\begin{aligned} I(X;Y) &= H(Y) - H(Y|X) \\ &= \tfrac{11}{4} - \tfrac{3}{4}\log_2 3 - \tfrac{3}{2} \\ &= \tfrac{5}{4} - \tfrac{3}{4}\log_2 3. \end{aligned}$$

This is approximately 0.061 bits per channel use. The output entropy includes uncertainty introduced by both errors and erasures. The answer is (a).

6. Now let $\Pr(X = 1) = q$. Which input probability maximizes $I(X;Y)$? Include a complete and convincing justification.

(a) $q = 0$.

(b) Every $q \in [0,1]$ gives the same mutual information.

(c) $q = \frac{1}{2}$. ⋆

> *Solution:* The output-law formula in Section 3.2 gives
>
> $$\Pr(Y = 0) = \tfrac{2-q}{4},$$
>
> while $\Pr(Y = 1) = \frac{1+q}{4}$ and $\Pr(Y = \star) = \frac{1}{4}$. Expanding the entropy sum as in the worked bit-erasing problem, we obtain
>
> $$H(Y) = h_2(\tfrac{1}{4}) + \tfrac{3}{4} h_2\big(\tfrac{1+q}{3}\big).$$
>
> Question 4 gives $H(Y|X) = \frac{3}{2}$ for every q. Hence
>
> $$I(X;Y) = h_2(\tfrac{1}{4}) + \tfrac{3}{4} h_2\big(\tfrac{1+q}{3}\big) - \tfrac{3}{2}.$$
>
> By Section 3.1, binary entropy is maximal when its argument is $\frac{1}{2}$. Solving $\frac{1+q}{3} = \frac{1}{2}$ gives $q = \frac{1}{2}$, with the maximum found in question 5. The answer is (c).

7. Compare this channel with the standard bit-erasing channel that erases with probability $\frac{1}{4}$ and otherwise outputs the transmitted bit unchanged, also with equiprobable input. Which conveys more information? Justify both information values and explain the difference.

(a) Both convey $\frac{3}{4}$ bit per use, since their erasure probabilities agree.

(b) The bit-erasing channel, which conveys $\frac{3}{4}$ bit per use. ⋆

(c) This channel, since a flipped bit still contributes one bit of information.

> *Solution:* The worked bit-erasing problem in Section 3.2 gives
>
> $$I_{\text{erasing}} = (1 - \tfrac{1}{4}) h_2(\tfrac{1}{2}) = \tfrac{3}{4}.$$
>
> For our channel, question 5 gives $\frac{5}{4} - \frac{3}{4}\log_2 3$. The difference is
>
> $$I_{\text{erasing}} - I(X;Y) = \tfrac{3}{4}\log_2 3 - \tfrac{1}{2} > 0,$$
>
> since $\log_2 3 > 1$. In the bit-erasing channel, every unerased output identifies the input. In our channel, an output bit may have been flipped, so uncertainty remains even when there is no erasure. The answer is (b).